CAMPUSBITE

Online Food Ordering System for College Canteen

E.D.G.E. 2026 - Project Draft Document

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1. Introduction

College canteens are important places where students, teaching staff, and other college employees buy snacks, meals, and drinks during the day. At busy times, many people may reach the counter together. This can create long queues, confusion while placing orders, and delays in receiving food. Students may also have limited time between lectures, practical sessions, and examinations.

CampusBite is a proposed online food ordering system designed for a college canteen. It will allow students to view available food items on a website, add items to a cart, place an order, and check the order status. Canteen staff will be able to view incoming orders and update their status as they prepare the food.

The project will use HTML and CSS to create the website pages, PHP to handle the application logic, and MySQL to store menu items and order information. XAMPP will provide a local development environment for running PHP and MySQL during development and demonstration.

The main idea is to make the ordering process more organised and convenient. Instead of depending only on a physical queue, students can use the website to select their food and submit their order. The system is intended as a student project prototype and can be improved further after feedback from students and canteen staff.

2. Problem Statement

In a traditional college canteen, students usually have to stand in a queue to see the menu, place an order, and wait for the staff to prepare it. During breaks and lunch hours, many students may arrive at the same time. This can lead to long waiting times and crowding near the counter.

Orders are often communicated verbally. In a noisy or crowded environment, staff may misunderstand an item, quantity, or special request. Students may also be unsure whether their order has been received or how long they need to wait. If the canteen menu or availability changes, it may be difficult for students to know about the change before reaching the counter.

These issues can reduce the amount of time students have for eating and returning to class. They can also make it harder for canteen staff to organise several orders at once. A simple digital system could help by displaying menu items clearly and recording each order in a structured format.

The problem addressed by CampusBite is the lack of a simple online method for students to browse a canteen menu, submit food orders, and check order progress. The proposed website aims to reduce confusion and improve order organisation. It will not guarantee zero waiting time, but it can make order details easier to manage.

3. Aim and Objectives

The main aim of CampusBite is to develop a simple, user-friendly website through which students can order food from their college canteen. The website will provide a digital menu, a shopping cart, order placement, and a basic order-status feature. It will also provide a staff page for viewing orders and updating their status.

The objectives of the project are:

- To display canteen food items with their names, prices, and availability information.
- To allow students to select food items and change quantities before placing an order.
- To calculate the order total and save order details in a database.
- To provide an order reference and a basic way to check order status.
- To allow canteen staff to view submitted orders and update their progress.
- To use HTML, CSS, PHP, and MySQL to build a working web-based prototype.
- To test the main functions and identify areas that need improvement.

These objectives will help the team build the project in manageable stages. The first version will focus on the core ordering process rather than advanced features such as online payment, delivery tracking, or mobile applications. The team will check whether each objective has been met through functional testing and demonstration.

4. Scope of the Project

The scope of CampusBite is to create a web-based prototype for ordering food from a college canteen. The expected users are students who place orders and authorised canteen staff who manage incoming orders. The prototype will be developed and tested in a local environment using XAMPP, PHP, and MySQL.

The student side will include viewing the menu, adding food items to a cart, changing quantities, submitting an order, and checking the status of an order using the available order details. The staff side will include viewing submitted orders and changing an order status, such as Pending, Preparing, or Ready.

The database will store menu information and order records. The project will include basic input validation and database queries designed to reduce common errors. The team will test the application using sample menu items and sample orders before presenting it.

The initial scope does not include real online payments, integration with college identity systems, automatic stock management, SMS notifications, or a live mobile application. These can be considered in future versions. The prototype is not intended for real public use until authentication, authorisation, security protections, privacy handling, and deployment configuration have been properly implemented and tested.

5. Proposed System

CampusBite proposes a website that connects students with the college canteen through a simple ordering process. Students will open the website, view the menu, choose items, add them to the cart, and submit the order. The system will calculate the total and save the order in the MySQL database.

After an order is submitted, the staff page will show the order details so that canteen staff can prepare the selected items. Staff can update the order status as work progresses. Students can then use the order lookup feature to check the latest status supported by the prototype.

The application will use HTML to structure the pages and CSS to style them. PHP will process user actions, manage sessions for the cart, validate submitted information, and communicate with MySQL. MySQL will store menu items, orders, and order-item details. XAMPP will be used to run Apache and MySQL on the development computer.

The proposed system aims to keep the interface easy to understand and the workflow short. The first version will use sample data and basic order lookup. Before real deployment, the system would need secure staff login, appropriate student verification, stronger access controls, protection against common web attacks, backups, and testing with the college canteen.

6. System Modules

CampusBite will be divided into modules so that each part can be developed and tested separately. This will also help group members share the work more easily.

1. Menu Module: Displays available food items and their prices. In a later version, staff may be able to add, edit, or remove menu items.
2. Cart Module: Allows students to add selected items, change quantities, remove items, and view the estimated total before submitting an order.
3. Order Placement Module: Receives the student's order details, validates the information, calculates the total, and stores the order in the database.
4. Order Tracking Module: Allows a student to look up an order and view its current status using the details supported by the prototype.
5. Staff Order Module: Displays incoming orders and allows staff to update the status as Pending, Preparing, or Ready.
6. Database Module: Stores menu information, order records, and the items included in each order. It helps keep the information organised and available to the relevant pages.

Each module will be tested individually and then tested as part of the complete workflow. Access to the staff module must be protected with authentication before the system is used beyond a classroom demonstration.

7. Technology Stack

HTML (HyperText Markup Language) will be used to create the structure of the web pages. It will define elements such as headings, forms, buttons, menu lists, tables, and order details. HTML provides the basic layout that the browser displays to the user.

CSS (Cascading Style Sheets) will be used to improve the appearance of the website. It will control colours, spacing, fonts, page layout, buttons, and responsive behaviour. The design will aim to be readable on common laptop and mobile screen sizes.

PHP will be used for server-side processing. It will handle form submissions, cart operations, order creation, order lookup, and staff status updates. Database queries should use prepared statements and input validation to reduce security risks.

MySQL will be used as the database management system. It will store food items, orders, and the details of items belonging to each order. Keeping this information in a database makes it easier to retrieve and update order records.

XAMPP will provide a local development setup containing Apache and MySQL/MariaDB services for running and testing the PHP application. Visual Studio Code or another code editor can be used to write the files. GitHub will be used for version control and collaboration, allowing group members to share code and track changes.

8. System Workflow

The proposed workflow begins when a student opens the CampusBite website. The home page displays the available food items and their prices. The student reviews the menu and selects the items they want to order.

Next, the selected items are added to the cart. The student checks the items, changes quantities if required, and reviews the total amount. Before the order is saved, the website should check that the required details are present and that the selected quantities are valid.

When the student submits the order, PHP processes the request and stores the order and its item details in the MySQL database. The system then provides an order reference or confirmation message, depending on the implemented version.

Canteen staff open the staff order page to view new orders. They prepare the food and update the order status as appropriate. The student can use the order lookup page to check the status. The workflow ends when the order is ready and collected by the student.

Simple workflow: Student opens website → Views menu → Adds items to cart → Confirms order → Order is saved → Staff prepares food → Staff updates status → Student checks status and collects order.

9. Basic System Design

The system will follow a simple three-part design. The presentation part is the website that users see in their browser. It is created mainly with HTML and CSS. The application part is written in PHP and handles the rules for adding items, submitting orders, calculating totals, and updating statuses. The data part is the MySQL database, which stores menu and order information.

At a high level, the student interacts with the web pages. The browser sends a request to the PHP application. PHP validates the request and communicates with the database when information must be read or saved. The application then returns a page showing the menu, cart, confirmation, or order status.

The database can be organised into related tables. A menu table stores food item details such as item ID, name, price, and availability. An orders table stores each order's reference, student identifier, total, status, and creation time. An order-items table records which menu items and quantities belong to each order. The final structure should be confirmed during implementation.

Basic design flow: Browser (HTML/CSS) ↔ PHP application ↔ MySQL database. The design separates the page layout, application logic, and stored data, making the project easier to maintain. Before real use, the design must include secure staff authentication, access restrictions, safe error handling, and appropriate protection of student information.

10. Testing Plan

Testing will be carried out to check whether CampusBite works as expected and whether common user mistakes are handled safely. The team will test each module separately and then test the complete ordering process from menu selection to order-status checking.

Menu testing will confirm that food items and prices appear correctly. Cart testing will check adding an item, increasing and decreasing quantities, removing items, and calculating the total. Order placement testing will check that valid orders are saved and that missing or invalid information is rejected with a clear message.

Database testing will confirm that order records and order-item details are saved consistently. Order tracking testing will check that the correct order status is shown. Staff module testing will confirm that status updates are saved and displayed correctly. The team should also test invalid order references and unexpected form inputs.

Testing will use sample data and should not include real student passwords or sensitive personal information. The team will record each test, the expected result, the actual result, and whether the test passed. Security testing and staff-login testing must be added before the system is considered for real use.

11. Expected Outcomes and Benefits

The expected outcome is a working prototype that demonstrates how a college canteen could receive and organise food orders through a website. Students should be able to browse the sample menu, add items to a cart, submit an order, and check the order status supported by the system. Staff should be able to view submitted orders and update their status in the demonstration.

The website may help reduce confusion caused by verbally communicating several orders at a busy counter. A structured order record can make it easier for staff to see selected items and quantities. Showing menu prices before an order is submitted can also help students review their choices and estimated total.

The project will provide practical learning for the team in front-end web design, server-side programming, database design, testing, and version control. Working with GitHub will help group members coordinate code changes and maintain a shared project history.

These benefits are expected rather than guaranteed. The actual effect on queues and waiting time would need to be measured through user feedback and testing in the real canteen. The first version is a prototype, so it should not be described as a fully deployed or production-ready service.

12. Future Enhancements

After the basic prototype is completed, CampusBite could be improved with additional features based on feedback from students and canteen staff. One important improvement would be secure login for staff and a suitable way to verify students. Role-based access would ensure that only authorised staff can view and update orders.

Online payment could be added through a trusted payment provider if the college approves it. This would require secure payment integration and careful handling of payment confirmation. The project should not store raw card details. Notifications through email or SMS could inform students when an order is accepted or ready.

Other possible improvements include an admin page for managing menu items, marking items as unavailable, daily sales reports, order history, estimated preparation times, and a responsive mobile-friendly interface. A QR code at the canteen could open the menu quickly. A later version could also support feedback and ratings.

Before adding advanced features, the team should improve security, validation, accessibility, database backups, and error handling. Future enhancements will be prioritised according to the time available, the needs of the college, and feedback gathered during testing.

13. Conclusion

CampusBite is a proposed online food ordering system for a college canteen. It addresses common difficulties in a manual ordering process, such as queues, unclear order details, and uncertainty about order progress. The project aims to provide a simple digital method for students to browse a menu, select food, submit orders, and check their status.

The system will be developed using HTML, CSS, PHP, and MySQL, with XAMPP used for local testing. Its main modules will include the food menu, cart, order placement, order tracking, staff order management, and database storage. The team will test the modules using sample data and document the results.

The project also gives the team an opportunity to practise web development, database management, teamwork, and version control through GitHub. A clear division of tasks and regular testing will help the group complete the prototype and prepare for the E.D.G.E. 2026 demonstration.

CampusBite will initially be a prototype, not a production-ready service. Secure staff authentication, appropriate access controls, and further testing will be necessary before any real canteen deployment. With these improvements, the project could be developed into a more complete and useful campus service.

14. Project Details and Team Responsibilities

Project title: CampusBite — Online Food Ordering System for College Canteen.

Competition: E.D.G.E. 2026.

Programme and institution: First-year M.Sc. Computer Science, MIT World Peace University, Pune.

Technologies: HTML, CSS, PHP, MySQL, XAMPP, and GitHub.

Team members: Add the full names and college details of all team members before submission.

Team leader: Add the name of the person responsible for coordinating the team and submitting the project draft and repository link.

Suggested task division: one member can work on the HTML/CSS pages; one can work on PHP form processing and cart features; one can work on the database and SQL; and one can coordinate testing, documentation, and GitHub. The team can adjust these responsibilities based on the number of members and their skills.

All members should understand the complete project, not only their own assigned part. The team should review the code together, test the main workflow, keep the GitHub repository organised, and make sure the project draft accurately describes the features that have actually been implemented.

Before submission, replace these placeholders with the correct team details and confirm that the GitHub repository contains the latest project files and a README with setup instructions.